Zaichen Hao

+1 224 844 1206 | felix772.github.io | felixhao772@gmail.com

EDUCATION

University of Florida — Gainesville, FL
BS in Computer Science, Minor in Mathematics — Expected May 2028
UF Distinguished Scholars Scholarship Recipient
Relevant Coursework: Operating Systems; Algorithms; Database & Information Systems; Computer Organization

Nanyang Technological University — Singapore
Study Abroad Program in Machine Learning and Artificial Intelligence — Jun 2025 - Aug 2025

WORK EXPERIENCE

Zhongguancun Institute of Artificial Intelligence — Beijing, China
AI Engineering Intern — May 2026 – Present

- Automated technical research workflows in Python using web scraping, API integration, data parsing, and metadata analysis to evaluate academic papers and technical resources for AI History Show, an interactive web exhibition for the 70th anniversary of the Dartmouth Conference.
- Analyzed open-source repositories using OScanner to assess code quality, commit patterns, contributor activity, dependency health, and potential AI-generated code risks.

PROJECTS

Order Matching Engine — Nov 2025

- Built a high-performance C++ order matching engine for a single equity, implementing price-time priority, deterministic execution, strict correctness guarantees, and integer-based price representation to avoid floating-point inaccuracies.
- Optimized latency-critical data paths using ordered price-level maps and unordered ID-location mapping, achieving ~2.2 µs effective per-order latency and ~450K orders/sec on a single core.
- Profiled hot paths with Google Benchmark and Linux perf to identify bottlenecks and improve consistency.

Earnings Call Intelligence Agent — Apr 2026

- Built an AI-powered earnings call intelligence backend with FastAPI, OpenAI API and PostgreSQL with pgvector, exposing 6 REST endpoints for transcript ingestion, RAG-based Q&A, and quarter-over-quarter financial analysis using chunked embeddings and top-k vector similarity search.
- Implemented a structured LLM report engine to compare current and prior-quarter transcripts and identify tone shifts, demand trends, margin changes, guidance updates, and risk flags.
- Added 31 unit tests across schema validation, chunking, health checks, and quarter-resolution logic while supporting 3 transcript ingestion paths.

Film Revenue Predictor — Jul 2025

- Led a 7-person AI course project at Nanyang Technological University, defining project scope and technical direction while building machine learning models with pandas and scikit-learn to predict film revenue from genre, budget, language, cast, and other 2019–2024 data features.
- Improved random forest model performance through k-fold cross-validation and hyperparameter tuning, increasing R^2 from 0.47 to 0.86.

ADDITIONAL

Technical Skills: C++, Python, SQL, Linux, Low-Latency Systems, Google Benchmark, Numerical Computing
Training: Selected as 1 of 8 Official Members from 100+ applicants for QiShi Club's Low Latency Systems course; delivered a lecture on instrumentation and performance optimization and two on data structures.
Publication: "Comparison Between Different Qubit Designs for Quantum Computing." *IJHSR*, 2024